

Benjamini-Hochberg FDR Correction — All Hypothesis Tests (Raw vs. Adjusted p-values, dashed line = significance threshold)

Hypothesis Test

E18: Reduction across families in Clifford+T (KW, H13)

E17: Reduction across topologies (KW, H12)

E16: Reduction across window sizes (KW, H11)

E15: Reduction across compilers (KW, H10)

E14: Reduction across families x optimizers (KW, H9)

E12: Reduction across optimization levels (KW, H8)

E11: Reduction across circuit families (KW, H7)

E10: Reduction across optimizers (Kruskal-Wallis)

E10: Reduction across circuit families (Kruskal-Wallis)

E5: Reduction across depths (Kruskal-Wallis)

E4: Runtime across optimizers (Kruskal-Wallis)

E4: Reduction across optimizers (Kruskal-Wallis)

E3: Reduction across qubit counts (Kruskal-Wallis)

E2: Entanglement-reduction correlation (Pearson)

E1: Reduction across depths (Kruskal-Wallis)

